

EXACT DISTRIBUTIONS ASSOCIATED WITH AN $h \times k$ CONTINGENCY TABLE

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Summary: In this paper a multidimensional extension of Appell's F_1 -hypergeometric function is defined as $F_{1;r,s}(\cdot)$ and applied as generating function of the joint distribution of the cell frequencies of an $h \times k$ contingency table. From the factorial moment generating function associated with $F_{1;r,s}(\cdot)$, marginal distributions of the frequencies in an $h \times k$ table are derived. The factorial moment generating functions associated with conditional distributions of the cell frequencies are written in terms of hypergeometric functions.

The approach presented here differs from that of Haldane (1940) and others in that an expression for the general factorial moment of any order, associated with a two-way table is derived.

The limiting form for the distribution generated by $F_{1;r,s}(\cdot)$ is shown to be multivariate normal and from this the necessary correction factor to Pearson's chi-square statistic is found.

To simplify the algebraic operations some mathematical properties of the $F_{1;r,s}(\cdot)$ function are mentioned. An integral and a system of differential equations for the $F_{1;r,s}(\cdot)$ function are presented.

1. INTRODUCTION

During the last two decades an overwhelming amount of literature on the analysis of categorical data has appeared. A contributing reason is the development of log-linear model analysis for analysing contingency tables by Goodman (1970), Haberman (1973a), Bishop, Fienberg and Holland (1975) and Gokhale and Kullback (1978) and others. Recently the analysis of categorical data arising from complex sampling schemes has been investigated by amongst others, Fellegi (1980) and Rao and Scott (1981, 1984). Aspects such as the transformation of contingency tables (Meyer (1982)), serial dependence in contingency tables (Tavare (1983)) and the use of prior information in the analysis of contingency tables (Albert and Gupta (1983)) have also received attention.

Notwithstanding the vast literature on the analysis of contingency tables, apart from Haldane (1940), Lancaster (1969), Zelen (1972) and Weis (1978) not much work has been done on the development of a comprehensive exact distribution theory regarding contingency tables. In this paper comprehensive exact distributional results are presented by showing that the results for an $h \times k$ contingency table are derived by considering a special case of a class of distributions represented by an extension of Appell's F_1 -hypergeometric function. This extension is denoted by $F_{1;r,s}(\cdot)$.

It is well known that the distribution associated with a 2×2 table is that of the usual ${}_2F_1(\cdot)$ hypergeometric function (see Lancaster (1969)). Steyn (1951, 1955, 1957) studied an extension of Appell's F_1 -hypergeometric function (see Erdélyi (1953, p. 224)) which in the notation to be introduced later can be denoted as $F_{1;1,(s-1)}(\cdot)$, in the context of $2 \times s$ contingency tables. The function $F_{1;r,s}(\cdot)$ introduced in this paper therefore constitutes a logical extension of the $F_{1;1,s}(\cdot)$ function in the distribution theory for two-way contingency tables. These results could

possibly be used to extend the Peizer approximations to the hypergeometric distribution (see Ling and Pratt (1984)).

Furthermore, from Metha and Patel (1983) and Pagano and Halvorsen (1981) it can be seen that there is still a strong interest in optimal computing procedures of the exact probabilities in $h \times k$ contingency tables. Dickey (1983) in the context of multiple hypergeometric functions, has emphasised the need for statisticians to be informed of developments in the mathematics of special functions.

In section 2 the exact probability distribution under the usual hypothesis of independent row and column classifications is derived from first principles and presented as a special case of the $F_{1;r,s}(\cdot)$ function. Haldane (1940) in deriving the mean and variance of Pearson's chi-square statistic develops individual expressions for the first four moments and remarks on the increasing complexity for the expressions involving higher order moments (p. 354). In this section an expression for the general factorial moment of any order associated with a two-way contingency table under the hypothesis of independence is derived by using an integral presentation for $F_{1;r,s}(\cdot)$.

Distributional results associated with marginal distributions of the $h \times k$ contingency table are given in section 3. In section 4 some distributional and regression results concerning conditional distributions for the $h \times k$ contingency table are presented while the limiting form of the distribution for large parameter values is considered in section 5.

The integral form, a system of differential equations associated with this $F_{1;r,s}(\cdot)$ function and some aspects regarding the derivation of the fmgf are given in separate appendices.

2. PROBABILITY DISTRIBUTION AND FACTORIAL MOMENT GENERATING FUNCTION

Consider the following $h \times k$ contingency table:

f_{11}	$f_{12} \dots f_{1k}$	$n_{1.}$
f_{21}	$f_{22} \dots f_{2k}$	$n_{2.}$
$\vdots$		
f_{h1}	$f_{h2} \dots f_{hk}$	$n_{h.}$
$n_{.1}$	$n_{.2} \dots n_{.k}$	n

in which:

$$f_{ik} = n_{i.} - \sum_{j=1}^{k-1} f_{ij}, \quad n_{h.} = n - \sum_{i=1}^{h-1} n_{i.}$$

$$(1) \quad f_{hj} = n_{.j} - \sum_{i=1}^{h-1} f_{ij}, \quad n_{.k} = n - \sum_{j=1}^{k-1} n_{.j}$$

$$\text{and} \quad f_{hk} = n_{h.} + n_{.k} - n + \sum_{i=1}^{h-1} \sum_{j=1}^{k-1} f_{ij}.$$

Given the usual assumption of an underlying multinomial sampling model and the condition of fixed marginal totals, the probability function in the random variables f_{ij} , $i=1,2,\dots,h-1$; $j=1,2,\dots,k-1$ with associated cell probabilities p_{ij} under the hypothesis H_0 of independent row and column classifications, is given by:

$$\begin{aligned} P_0 &= P(\{f_{ij}\} | H_0 \text{ and fixed } n_{i.}, n_{.j}) \\ &= \frac{\phi_1(f_{11}, f_{12}, \dots, f_{hk})}{\phi_2(n_{1.}, n_{2.}, \dots, n_{h.}) \phi_3(n_{.1}, n_{.2}, \dots, n_{.k})}, \end{aligned}$$

$$\text{where } \phi_1(\cdot) = n! \prod_{i=1}^h \prod_{j=1}^k (p_{ij}^{f_{ij}} / f_{ij}!)$$

$$\phi_2(\cdot) = n! \prod_{i=1}^h (p_{i\cdot}^{n_{i\cdot}} / n_{i\cdot}!)$$

$$\phi_3(\cdot) = n! \prod_{j=1}^k (p_{\cdot j}^{n_{\cdot j}} / n_{\cdot j}!)$$

$$\text{and } p_{i\cdot} = \sum_{j=1}^k p_{ij}, \quad p_{\cdot j} = \sum_{i=1}^h p_{ij} \quad \text{and where, under the}$$

hypothesis of independence, $p_{ij} = p_{i\cdot} p_{\cdot j}$. Thus the above conditional probability function reduces to

$$(2) \quad P_0 = \frac{\prod_{i=1}^h n_{i\cdot}! \prod_{j=1}^k n_{\cdot j}!}{n! \prod_{i=1}^h \prod_{j=1}^k f_{ij}!}$$

Using (1), it follows that (2) can be written in the form of an $(h-1)(k-1)$ multivariate factorial function:

$$\frac{n_{h\cdot}! n_{\cdot k}! \prod_{i=1}^{h-1} n_{i\cdot}! \prod_{j=1}^{k-1} n_{\cdot j}!}{(n_{h\cdot} + n_{\cdot k} - n)! n! (n_{h\cdot} + n_{\cdot k} - n + \sum_{i=1}^{h-1} \sum_{j=1}^{k-1} f_{ij}) \prod_{i=1}^{h-1} \prod_{j=1}^{k-1} 1/f_{ij}!}$$

where $a^{(r)} = a(a-1) \dots (a-r+1)$, provided that $n_{h\cdot} + n_{\cdot k} - n \geq 0$, so that the denominator remains positive. Under this condition the probability generating function (pgf) associated with (2) is thus given by:

$$(3) \frac{n_h! n_k!}{(n_h + n_k - n)! n!} \sum_{f_{11}} \sum_{f_{12}} \dots \sum_{f_{(h-1)(k-1)}}$$

$$\frac{\prod_{i=1}^{h-1} n_i! \left(\sum_{j=1}^{k-1} f_{ij} \right)!}{1} \frac{\prod_{j=1}^{k-1} n_j! \left(\sum_{i=1}^{h-1} f_{ij} \right)!}{1} \frac{\prod_{i=1}^{h-1} \prod_{j=1}^{k-1} f_{ij}!}{\prod_{i=1}^{h-1} \prod_{j=1}^{k-1} t_{ij} / f_{ij}!}$$

$$(n_h + n_k - n + \sum_{i=1}^{h-1} \sum_{j=1}^{k-1} f_{ij})!$$

Apart from the constant, (3) is obtained as a special case of the following extension of Appell's F_1 -hypergeometric function defined as:

$$F_{1;r,s}(a_1, a_2, \dots, a_r; b_1, b_2, \dots, b_s; c; t_{11}, t_{12}, \dots, t_{rs})$$

$$(4) = \sum_{x_{11}} \sum_{x_{12}} \dots \sum_{x_{rs}} \frac{\prod_{i=1}^r \frac{[\sum_{j=1}^s x_{ij}]}{a_i} \prod_{j=1}^s \frac{[\sum_{i=1}^r x_{ij}]}{b_j}}{c^{\sum_{i=1}^r \sum_{j=1}^s x_{ij}}} \prod_{i=1}^r \prod_{j=1}^s \frac{x_{ij}}{t_{ij} / x_{ij}!}$$

where $a^{[r]} = a(a+1) \dots (a+r-1)$.

If $a_i = -n_i$, $b_j = -n_j$ and $c = n_h + n_k - n + 1$

then (4) yields, apart from the constant, the expression for the pgf associated with the $h \times k$ table given in (3), which is thus given by:

$$(5) C_1 F_{1;h-1,k-1}(-n_1, -n_2, \dots, -n_{(h-1)}; -n_1, -n_2, \dots, -n_{(k-1)};$$

$$n_h + n_k - n + 1; t_{11}, t_{12}, \dots, t_{(h-1)(k-1)})$$

where $C_1 = n_h! n_k! / (n_h + n_k - n)! n!$

Some mathematical properties of (4) are discussed in the appendices.

From the integral for (4) (see appendix A.1) given by:

$$\frac{\Gamma(c)}{\prod_{j=1}^s \Gamma(b_j) \Gamma(c - \sum_{j=1}^s b_j)} \int_{u_1 \geq 0} \int_{u_2 \geq 0} \dots \int_{u_s \geq 0} \prod_{j=1}^s u_j^{b_j-1} (1 - \sum_{j=1}^s u_j)^{c - \sum_{j=1}^s b_j - 1} du_1 du_2 \dots du_s,$$

$$\int_{u_1 \geq 0} \int_{u_2 \geq 0} \dots \int_{u_s \geq 0} \prod_{j=1}^s u_j^{b_j-1} (1 - \sum_{j=1}^s u_j)^{c - \sum_{j=1}^s b_j - 1} du_1 du_2 \dots du_s, \text{ where } \sum_{j=1}^s u_j \leq 1,$$

it follows that:

$$F_{1;r,s}(a_1, a_2, \dots, a_r; b_1, b_2, \dots, b_s; c; 1, 1, \dots, 1)$$

$$= \frac{\Gamma(c)}{\prod_{j=1}^s \Gamma(b_j) \Gamma(c - \sum_{j=1}^s b_j)} \int_{u_1 \geq 0} \int_{u_2 \geq 0} \dots \int_{u_s \geq 0} \left(\prod_{j=1}^s u_j^{b_j-1} \right) (1 - \sum_{j=1}^s u_j)^{c - \sum_{j=1}^s b_j - 1} du_1 du_2 \dots du_s$$

$$du_1 du_2 \dots du_s$$

$$(6) = \frac{\Gamma(c)}{\prod_{j=1}^s \Gamma(b_j) \Gamma(c - \sum_{j=1}^s b_j)} \frac{1}{\Gamma(\sum_{j=1}^s b_j)} \int_0^1 (1-v)^{c - \sum_{j=1}^s a_j - \sum_{j=1}^s b_j - 1} v^{\sum_{j=1}^s b_j - 1} dv \quad \begin{matrix} \text{(using Dirichlet's integral)} \\ \text{(see appendix A.1)} \end{matrix}$$

$$= {}_2F_1\left(\sum_{i=1}^r a_i; \sum_{j=1}^s b_j; c; 1\right)$$

$$= \Gamma(c) \Gamma(c - \sum_{i=1}^r a_i - \sum_{j=1}^s b_j) / \Gamma(c - \sum_{i=1}^r a_i) \Gamma(c - \sum_{j=1}^s b_j)$$

$$\text{where } {}_2F_1(a; b; c; t) = \sum_x \frac{a^{[x]} b^{[x]} / c^{[x]}}{x!} \cdot t^x / x!$$

The factorial moment generating function $M(\alpha_{11}, \alpha_{12}, \dots, \alpha_{rs})$ of (4) is obtained by setting $t_{ij} = 1 + \alpha_{ij}$; $i=1, 2, \dots, r$; $j=1, 2, \dots, s$, so that

$$(7) M(\alpha_{11}, \alpha_{12}, \dots, \alpha_{rs}) = {}_2F_{1;r,s}(a_1, a_2, \dots, a_r; b_1, b_2, \dots, b_s; c; 1 + \alpha_{11}, 1 + \alpha_{12}, \dots, 1 + \alpha_{rs})$$

where, remembering the fact that $M(\cdot)$ is a fmgf,

$$C_2^{-1} = {}_1F_{1;r,s}(a_1, a_2, \dots, a_r; b_1, b_2, \dots, b_s; c; 1, 1, \dots, 1)$$

$$\begin{aligned} & \frac{\Gamma(c) \Gamma(c - \sum_{i=1}^r a_i - \sum_{j=1}^s b_j)}{\Gamma(c - \sum_{i=1}^r a_i) \Gamma(c - \sum_{j=1}^s b_j)} \\ &= \frac{\Gamma(c) \Gamma(c - \sum_{i=1}^r a_i - \sum_{j=1}^s b_j)}{\Gamma(c - \sum_{i=1}^r a_i) \Gamma(c - \sum_{j=1}^s b_j)} \end{aligned}$$

since $M(\alpha_{11}, \alpha_{12}, \dots, \alpha_{rs}) = 1 + \text{terms containing } \alpha_{11}, \alpha_{12}, \dots, \alpha_{rs}$.

In integral form (7) is given by:

$$M(\alpha_{11}, \alpha_{12}, \dots, \alpha_{rs}) = C_2 \frac{\Gamma(c)}{\prod_{j=1}^s \Gamma(b_j) \Gamma(c - \sum_{j=1}^s b_j)} \int_0^1 \int_0^1 \dots \int_0^1 \prod_{j=1}^s u_j^{b_j-1} \dots$$

$$\begin{aligned} & \frac{\Gamma(c - \sum_{i=1}^r a_i)}{\prod_{j=1}^s \Gamma(b_j) \Gamma(c - \sum_{i=1}^r a_i - \sum_{j=1}^s b_j)} \int_0^1 \int_0^1 \dots \int_0^1 \prod_{j=1}^s u_j^{b_j-1} (1 - \sum_{j=1}^s u_j)^{c - \sum_{i=1}^r a_i - \sum_{j=1}^s b_j - 1} \\ & \quad \prod_{i=1}^r (1 - \sum_{j=1}^s u_j)^{c - \sum_{j=1}^s b_j - 1} \prod_{i=1}^r (1 - \sum_{j=1}^s u_j (1 + \alpha_{ij}))^{-a_i} du_1 du_2 \dots du_s \\ &= \frac{\Gamma(c - \sum_{i=1}^r a_i)}{\prod_{j=1}^s \Gamma(b_j) \Gamma(c - \sum_{i=1}^r a_i - \sum_{j=1}^s b_j)} \int_0^1 \int_0^1 \dots \int_0^1 \prod_{j=1}^s u_j^{b_j-1} (1 - \sum_{j=1}^s u_j)^{c - \sum_{i=1}^r a_i - \sum_{j=1}^s b_j - 1} \\ & \quad \prod_{i=1}^r (1 - \sum_{j=1}^s \frac{u_j}{1 - \sum_{j=1}^s u_j} \alpha_{ij})^{-a_i} du_1 du_2 \dots du_s, \text{ where } \sum_{j=1}^s u_j \leq 1. \end{aligned}$$

Expanding the factors of the last product as power series in the α_{ij} ,

$$M(\alpha_{11}, \alpha_{12}, \dots, \alpha_{rs})$$

$$= \frac{\Gamma(c - \sum_{i=1}^r a_i)}{\prod_{j=1}^s \Gamma(b_j) \Gamma(c - \sum_{i=1}^r a_i - \sum_{j=1}^s b_j)} \sum_{y_{11}} \sum_{y_{12}} \dots \sum_{y_{rs}} \prod_{i=1}^r (a_i)^{[\sum_{j=1}^s y_{ij}]} \prod_{j=1}^s (\alpha_{ij}^{y_{ij}} / y_{ij}!) \cdot$$

$$\int_0^1 \dots \int_0^1 \prod_{j=1}^s u_j^{b_j + \sum_{i=1}^r y_{ij} - 1} (1 - \sum_{j=1}^s u_j)^{c - \sum_{i=1}^r a_i - \sum_{j=1}^s b_j - \sum_{i=1}^r \sum_{j=1}^s y_{ij} - 1} du_1 du_2 \dots du_s$$

and writing the integral in terms of gamma functions as in (6) above, it follows directly that $M(\cdot)$ reduces to

$$M(\alpha_{11}, \alpha_{12}, \dots, \alpha_{rs})$$

$$= \sum_{y_{11}} \sum_{y_{12}} \dots \sum_{y_{rs}} \frac{\Gamma(c - \sum_{i=1}^r a_i - \sum_{j=1}^s b_j - \sum_{i=1}^r \sum_{j=1}^s y_{ij})}{\Gamma(c - \sum_{i=1}^r a_i - \sum_{j=1}^s b_j)} \prod_{j=1}^s \frac{\Gamma(b_j + \sum_{i=1}^r y_{ij})}{\Gamma(b_j)} \prod_{i=1}^r (a_i)^{[\sum_{j=1}^s y_{ij}]} \cdot$$

$$\prod_{i=1}^r \prod_{j=1}^s \alpha_{ij}^{y_{ij}} / y_{ij}!$$

$$= \sum_{y_{11}} \sum_{y_{12}} \dots \sum_{y_{rs}} \frac{\prod_{i=1}^r (a_i)^{[\sum_{j=1}^s y_{ij}]} \prod_{j=1}^s (b_j)^{[\sum_{i=1}^r y_{ij}]} \prod_{i=1}^r \prod_{j=1}^s (\alpha_{ij})^{y_{ij}} / y_{ij}!}{(\sum_{i=1}^r a_i + \sum_{j=1}^s b_j - c + \sum_{i=1}^r \sum_{j=1}^s y_{ij})^{(\sum_{i=1}^r \sum_{j=1}^s y_{ij})}}$$

$$(8) = F_{1;r,s}(a_1, a_2, \dots, a_r; b_1, b_2, \dots, b_s; \sum_{i=1}^r a_i + \sum_{j=1}^s b_j - c + 1; -\alpha_{11}, -\alpha_{12}, \dots, -\alpha_{rs}).$$

The fmgf associated with an $h \times k$ table (i.e. associated with the pgf (5)) is obtained by setting

$$a_i = -n_{i.}, \quad b_j = -n_{.j} \quad \text{and} \quad c = n_{h.} + n_{.k} - n + 1$$

and from (8) is given by

$$(9) F(-n_1, -n_2, \dots, -n_{(h-1)}, -n_1, -n_2, \dots, -n_{(k-1)}; -\alpha_{11}, -\alpha_{12}, \dots, -\alpha_{(h-1)(k-1)}).$$

Some comments on (2) to (9) are necessary:

- (i) The expression (5) is a multivariate polynomial and the fmgf (9) can also be derived from (5) by algebraic operations. The method is similar to that used in appendix A.3.
- (ii) The integral used to obtain expression (8) from (7) assumes that the parameters $b_1, b_2, \dots, b_s$ are not zero or negative integers. That the transformation from (7) to (8) can be applied to transform (5) to (9) can readily be shown by either using the corresponding contour integral or by noting that the converging series (7) and (8) are both continuous functions in the parameters $b_1, b_2, \dots, b_s$ in a region around the relevant values in (5).
- (iii) The series (9) terminates before the denominator becomes zero. If the condition $n_h + n_k - n \geq 0$ is not satisfied (9) is still the fmgf of (2). This is proved in appendix A.3.

From (9) and the well-known relations between factorial and central moments, the first and second order moments for an $h \times k$ table are derived as (see also Haldane (1940)):

$$\begin{aligned} E(f_{ij}) &= n_i \cdot n_j / n, \quad i=1, 2, \dots, h-1; \quad j=1, 2, \dots, k-1, \\ &= n \hat{p}_i \cdot \hat{p}_j, \quad \text{where } \hat{p}_i = n_i / n \text{ and } \hat{p}_j = n_j / n \end{aligned}$$

$$\begin{aligned} \text{Var}(f_{ij}) &= (n - n_i) n_i \cdot n_j (n - n_j) / (n^2) \\ &= \left(\frac{n}{n-1} \right) n \hat{p}_i \cdot \hat{q}_i \cdot \hat{p}_j \cdot \hat{q}_j \quad \text{where } \hat{q}_i = 1 - \hat{p}_i, \\ &\quad \hat{q}_j = 1 - \hat{p}_j \end{aligned}$$

$$\begin{aligned} \text{Cov}(f_{ij}, f_{is}) &= -n_{.j} n_{.s} n_i / (n(n-1)), \quad s \neq j \\ &= -\left(\frac{n}{n-1}\right) \hat{p}_{.j} \hat{p}_{.s} \hat{p}_i. \end{aligned}$$

$$\begin{aligned} \text{Cov}(f_{ij}, f_{rj}) &= -n_i n_r n_{.j} / (n(n-1)), \quad r \neq i \\ &= -\left(\frac{n}{n-1}\right) \hat{p}_i \hat{p}_r \hat{p}_{.j}. \end{aligned}$$

$$\begin{aligned} \text{Cov}(f_{ij}, f_{rs}) &= + n_i n_r n_{.j} n_{.s} / (n(n-1)), \quad \begin{cases} s \neq j \\ r \neq i \end{cases} \\ &= +\left(\frac{n}{n-1}\right) \hat{p}_i \hat{p}_r \hat{p}_{.j} \hat{p}_{.s}. \end{aligned}$$

3. MARGINAL DISTRIBUTIONS

The fmgf of the marginal distribution of a single cell frequency $f_{\ell m}$ is obtained from (9) by substituting $\alpha_{ij} = 0$ for all $(ij) \neq (\ell m)$ and is given by:

$$\begin{aligned} F_{1;h-1,k-1}(-n_{.1}, -n_{.2}, \dots, -n_{(h-1)}; -n_{.1}, -n_{.2}, \dots, -n_{(k-1)}; -n; \\ 0, 0, \dots, -\alpha_{\ell m}, 0, 0, \dots, 0) = {}_2F_1(-n_{.}; -n_{.m}; -n; -\alpha_{\ell m}). \end{aligned}$$

The marginal distribution of $f_{\ell m}$ under the hypothesis of independence is therefore given by the distribution of $f_{\ell m}$ in the 2×2 table obtained by suitably aggregating over all cell frequencies except $f_{\ell m}$.

Similarly the fmgf of the marginal distribution of the ℓ -th row is given by:

$$F_{1,1,k-1}(-n_{\ell}; -n_{.1}, -n_{.2}, \dots, -n_{(k-1)}; -n; -\alpha_{\ell 1}, -\alpha_{\ell 2}, \dots, -\alpha_{\ell(k-1)}),$$

which is the fmgf of a $2 \times k$ contingency table.

The marginal distribution of $f_{\ell m}$ and f_{rs} where $\ell \neq r$ and $m \neq s$, i.e. the frequencies are in different rows and columns,

is obtained by a similar argument and is given by the distribution associated with a 3×3 table obtained by firstly collapsing over all rows except the ℓ -th and r -th and then over all columns except the m -th and s -th.

The pgf associated with this 3×3 table is, apart from a constant given by:

$$(10) \quad F_{1;2,2}(-n_{\ell}, -n_r, -n_m, -n_s; n-n_{\ell}, -n_r, -n_m, -n_s+1; t_{\ell m}, t_{rm}, t_{\ell s}, t_{rs}).$$

Setting $t_{rm} = t_{\ell s} = 1$ in (10) the pgf of the marginal distribution of $f_{\ell m}$ and f_{rs} is:

$$(11) \quad \sum_{f_{\ell m}} \sum_{f_{rs}} \frac{\binom{f_{\ell m}}{n_{\ell}} \binom{f_{rs}}{n_r} \binom{f_{\ell m}}{n_m} \binom{f_{rs}}{n_s}}{(n-n_{\ell}-n_r-n_m-n_s+f_{\ell m}+f_{rs}) \binom{f_{\ell m}+f_{rs}}{f_{\ell m}} t_{\ell m}^{f_{\ell m}/f_{\ell m}} t_{rs}^{f_{rs}/f_{rs}}},$$

$$F_3(-n_{\ell}+f_{\ell m}, -n_r+f_{rs}, -n_m+f_{\ell m}, -n_s+f_{rs}; n-n_{\ell}-n_r-n_m-n_s+f_{\ell m}+f_{rs}+1; 1, 1),$$

where $F_3(\cdot)$ represents Appell's well-known F_3 -hypergeometric function (Erdelyi (1953)) defined by:

$$(12) \quad F_3(\alpha, \alpha'; \beta, \beta'; \delta; t, u) = \sum_{i,j} \frac{\alpha[i]_{\alpha} [j]_{\beta} [i]_{\beta} [j]_{\delta}}{\delta[i+j]} \frac{t^i u^j}{i! j!}.$$

4. CONDITIONAL DISTRIBUTIONS AND REGRESSION PROPERTIES

The conditional distribution of $f_{\ell m}$ for given $f_{ij} = n_{ij} (ij \neq \ell m)$ is given by the distribution associated with the following 2×2 table:

$f_{\ell m}$	$n_{\ell.} - \sum_{j \neq m}^{k-1} n_{\ell j} - f_{\ell m}$	$n_{\ell.} - \sum_{j \neq m}^{k-1} n_{\ell j}$
$n_{.m} - \sum_{i \neq \ell}^{h-1} n_{im} - f_{\ell m}$	A	$n' - (n_{\ell.} - \sum_{j \neq m}^{k-1} n_{\ell j})$
$n_{.m} - \sum_{i \neq \ell}^{h-1} n_{im}$	$n' - (n_{.m} - \sum_{i \neq \ell}^{h-1} n_{im})$	n'

$$\text{where } n' = n_{h.} + n_{.k} + n_{\ell.} + n_{.m} - n + \sum_{i \neq \ell}^{h-1} \sum_{j \neq m}^{k-1} n_{ij}$$

$$= f_{\ell m} + n_{hk} + n_{\ell k} + n_{hm}$$

$$\text{and } A = n' - n_{\ell.} - n_{.m} + \sum_{i \neq \ell}^{h-1} n_{im} + \sum_{j \neq m}^{k-1} n_{\ell j} + f_{\ell m}.$$

Since the 2×2 table with fixed marginal totals leads to a univariate hypergeometric distribution the regression and scedastic equations can be written down as the mean and variance of the relevant hypergeometric distribution.

The regression value of $f_{\ell m}$ is thus given by the non-linear form:

$$(13) \quad \hat{f}_{\ell m} = \frac{(n_{\ell.} - \sum_{j \neq m}^{k-1} n_{\ell j})(n_{.m} - \sum_{i \neq \ell}^{h-1} n_{im})}{n_{h.} + n_{.k} + n_{\ell.} + n_{.m} - n + \sum_{i \neq \ell}^{h-1} \sum_{j \neq m}^{k-1} n_{ij}}.$$

The fmgf of the conditional distribution of the ℓ -th row of cell frequencies given $f_{ij} = n_{ij}$ for $i \neq \ell$ can be written down from (9) for a $2 \times k$ table as:

$$F_{1;1,k-1}(-n_{\ell.}; -(n_{.1} - \sum_{i \neq \ell}^{h-1} n_{i1}), -(n_{.2} - \sum_{i \neq \ell}^{h-1} n_{i2}), \dots, -(n_{.(k-1)} - \sum_{i \neq \ell}^{h-1} n_{i(k-1)});$$

$$-n''; -\alpha_1, -\alpha_2, \dots, -\alpha_{(k-1)}) \text{ where } n'' = n_{h.} + n_{\ell.}$$

This $F_{1;1,k-1}(\cdot)$ series was studied by Steyn (1957). The regression of $f_{\ell m}$ with respect to the other variables is of course also given by (13).

The conditional distribution of the four frequencies $f_{\ell m}$, $f_{\ell s}$, f_{rm} , f_{rs} given $f_{uv} = n_{uv}$, $(uv) \neq (\ell m)$, (ℓs) , (rm) and (rs) , is the same as the distribution associated with a 3×3 table obtained by subtracting the relevant given frequencies from the row and column totals and is, therefore, given by (10) but with $n_{\ell.}$, $n_{r.}$, $n_{.m}$, $n_{.s}$, n replaced respectively by:

$$n_{\ell.} = \sum_{j \neq m, s}^{k-1} n_{\ell j}, \quad n_{r.} = \sum_{j \neq m, s}^{k-1} n_{rj}, \quad n_{.m} = \sum_{i \neq \ell, r}^{h-1} n_{im}, \quad n_{.s} = \sum_{i \neq \ell, r}^{h-1} n_{is},$$

$$n_{h.} + n_{.k} + n_{\ell.} + n_{r.} + n_{.m} + n_{.s} - n + \sum_{i \neq \ell, r}^{h-1} \sum_{j \neq m, s}^{k-1} n_{ij}.$$

This indicates that the conditional distribution of $f_{\ell m}$, f_{rs} , i.e. two frequencies not in the same row and column, can be written down by expanding the amended form of (10) and is given by:

$$C_3 F_3(-n_{\ell.} + \sum_{j \neq m}^{k-1} n_{\ell j}, -n_{r.} + \sum_{j \neq s}^{k-1} n_{rj}, -n_{.m} + \sum_{i \neq \ell}^{h-1} n_{im}, -n_{.s} + \sum_{i \neq r}^{h-1} n_{is};$$

$$n''' - n_{\ell.} + \sum_{j \neq m}^{k-1} n_{\ell j} - n_{r.} + \sum_{j \neq s}^{k-1} n_{rj} - n_{.m} + \sum_{i \neq \ell, r}^{h-1} n_{im} - n_{.s} + \sum_{i \neq \ell, r}^{h-1} n_{is} + 1; t_{\ell m}, t_{rs})$$

where $n''' = n_{h.} + n_{.k} + n_{\ell.} + n_{r.} + n_{.m} + n_{.s} - n + \sum_{i \neq \ell, r}^{h-1} \sum_{j \neq m, s}^{k-1} n_{ij}$

and

$$C_3 = \frac{C_1 (n_{\ell.} - \sum_{j \neq m, s}^{k-1} n_{\ell j})^{(n_{\ell s})} (n_{r.} - \sum_{j \neq m, s}^{k-1} n_{rj})^{(n_{rm})} (n_{.m} - \sum_{i \neq \ell, r}^{h-1} n_{im})^{(n_{rm})} (n_{.s} - \sum_{i \neq \ell, r}^{h-1} n_{is})^{(n_{\ell s})}}{(n''' - n_{\ell.} + \sum_{j \neq m}^{k-1} n_{\ell j} - n_{r.} + \sum_{j \neq s}^{k-1} n_{rj} - n_{.m} + \sum_{i \neq \ell, r}^{h-1} n_{im} - n_{.s} + \sum_{i \neq \ell, r}^{h-1} n_{is})^{(n_{\ell s} + n_{rm})} n_{\ell s}! n_{rm}!}$$

The moments of this conditional distribution can be derived from the differential equations for the F_3 -hypergeometric function or by direct differentiation of the moment generating function. In either case the expressions for the moments contain proportions and/or products of the F_3 -function. (See Steyn and Stumpf (1983).)

Steyn (1957) derived the following regression equation for x on y for the probability function $\phi(x,y)$ generated by $CF_3(\cdot)$ where $F_3(\cdot)$ is given by (12):

$$\hat{x} = \alpha\beta / (\delta - \alpha - \beta - 1 + y).$$

Applying this result to the conditional distribution of $f_{\ell m}$ and f_{rs} , $\ell \neq r$, $m \neq s$, given $f_{\ell s} = n_{\ell s}$ and $f_{rm} = n_{rm}$ and $f_{uv} = n_{uv}$, $(uv) \neq (\ell m) \neq (rs)$ the following non-linear regression value for $f_{\ell m}$ is obtained:

$$(14) \quad \hat{f}_{\ell m} = \frac{(n_{\ell \cdot} - \sum_{j \neq m}^{k-1} n_{\ell j})(n_{\cdot m} - \sum_{i \neq \ell}^{h-1} n_{im})}{n_{h \cdot} + n_{\cdot k} + n_{\ell \cdot} + n_{\cdot m} - n + \sum_{i \neq \ell}^{h-1} \sum_{j \neq m}^{k-1} n_{ij}}$$

which is the same as (13) as could be expected.

5. LIMITING FORM FOR LARGE PARAMETER VALUES

That the limiting form of (2) is multivariate normal follows from the fact that the limiting form of the distribution associated with the $h \times k$ table with fixed marginal totals may be viewed as the resulting conditional multivariate normal limit of a multinomial distribution in $hk - 1$ random variables under $(h-1) + (k-1)$ linear relations between the variables imposed by the fixed marginal totals. The limiting form of (2) will, therefore, be an $(h-1)(k-1)$ -variate normal probability function.

Assuming that all the parameters of (4) are of the same order it can be shown that all probability functions generated by (4) will tend to the multivariate normal for large values of the parameters. As the algebraic operations are fairly straightforward only the method will be indicated here.

Let $t_{\ell m}$ be substituted with $e^{\beta_{\ell m}}$ (for all values of ℓ and m) in the system of differential equations satisfied by $F_{1;r,s}(\cdot)$ and which is given in appendix A.2, and let in the new system of differential equations thus obtained for the moment generating function $M(\cdot)$, $M(\cdot)$ be substituted with $e^{L(\cdot)}$, then the resulting cumulant generating function can be shown, using straightforward methods, to satisfy a set of equations which for $\beta = 0$ (for all ℓ, m) lead to recurrence formulae in cumulants, from which it follows that all cumulants are of the same order in the parameters. In standard units all cumulants of the third and higher order will therefore tend to zero when the parameters tend to infinity, so that the limiting form will be the multivariate normal probability function.

For an $h \times k$ contingency table the well-known Pearson chi-square statistic for the test of independent row and column classifications is given by:

$$\chi^2 = \underline{u}' \underline{A} \underline{u}$$

where $\underline{u}_{h \times 1} = \{(f_{ij} - n\hat{p}_i \cdot \hat{p}_j)\}$

and $\underline{A}_{(hk) \times (hk)} = (1/n)B = (1/n)(\text{Diag}\{1/\hat{p}_i \cdot \hat{p}_j\})$.

From the foregoing $\underline{u}$ is $SN(\underline{0}, V)$ (SN = singular normal) where the covariance matrix V can be written as:

$$V = (n^2/(n-1))(\underline{D}_i - \hat{p}_i \hat{p}_i') \otimes (\underline{D}_j - \hat{p}_j \hat{p}_j')$$

where, $D_i = \text{Diag}\{\hat{p}_{i.}\}$ and $\hat{p}_i = \{\hat{p}_{i.}\}$ (similarly for D_j and $\hat{p}_j$ in terms of $\hat{p}_{.j}$) and $\otimes$ indicates the Kronecker product.

Now, $X^2 = \underline{u}'\underline{A}\underline{u}$ will be distributed according to $\chi^2_{\text{tr}(\underline{A}\underline{V})}$ if and only if $\underline{V}\underline{A}\underline{V}\underline{A}\underline{V} = \underline{V}\underline{A}\underline{V}$ (see e.g. Searle (1971), p. 69).

From some simple matrix algebra it follows that

$((D_i - \hat{p}_i \hat{p}_i') \otimes (D_j - \hat{p}_j \hat{p}_j'))B$ is idempotent.

Since $\underline{V}\underline{A} = (n/(n-1))((D_i - \hat{p}_i \hat{p}_i') \otimes (D_j - \hat{p}_j \hat{p}_j'))B$, so that $((\frac{n-1}{n})\underline{V}\underline{A}\underline{V}\underline{A}\underline{V} = \underline{V}\underline{A}\underline{V}$ it follows that $((n-1)/n)X^2$ will have a central chi-square distribution with $\text{tr}(((D_i - \hat{p}_i \hat{p}_i') \otimes (D_j - \hat{p}_j \hat{p}_j'))B) = (h-1)(k-1)$ degrees of freedom.

6. OTHER DISTRIBUTIONS GENERATED BY $F_{1;r,s}(\cdot)$

The fact that the series $F_{1;r,s}(\cdot)$ gives a generalization of the series

$$F_{1;1,k}(\cdot) = \sum_{x=0}^{\infty} \frac{a[x]}{c[x]} \sum_{x_i=x}^k \prod_{i=1}^k b_i^{[x_i]} t^{x_i} / x_i! ,$$

which was used by Steyn (1951, 1955) and other writers as generating function of a wide class of discrete multivariate distributions suggests that $F_{1;r,s}(\cdot)$ will most probably also be of value for wider use.

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APPENDICES

A.1 INTEGRAL FORM FOR $F_{1;r,s}(\cdot)$

Consider the integral

$$\frac{\Gamma(c)}{\prod_{j=1}^s \Gamma(b_j) \Gamma(c - \sum_{j=1}^s b_j)} \int_{u_1 \geq 0} \int_{u_2 \geq 0} \dots \int_{u_s \geq 0} \prod_{j=1}^s u_j^{b_j-1} (1 - \sum_{j=1}^s u_j)^{c - \sum_{j=1}^s b_j-1} \\ \prod_{i=1}^r (1 - \sum_{j=1}^s u_j t_{ij})^{-a_i} du_1 du_2 \dots du_s$$

where $b_j > 0$, $j=1, 2, \dots, s$; $c - \sum_{j=1}^s b_j > 0$ and $\sum_{j=1}^s u_j \leq 1$.

This integral can be written as:

$$(15) \sum_{x_{11}} \sum_{x_{12}} \dots \sum_{x_{rs}} \frac{\Gamma(c)}{\prod_{j=1}^s \Gamma(b_j) \Gamma(c - \sum_{j=1}^s b_j)} \int_{u_1} \int_{u_2} \dots \int_{u_s} \left\{ \prod_{j=1}^s u_j^{b_j + \sum_{i=1}^r x_{ij} - 1} \right. \\ \left. (1 - \sum_{j=1}^s u_j)^{c - \sum_{j=1}^s b_j - 1} \prod_{i=1}^r a_i^{(\sum_{j=1}^s x_{ij})} \prod_{i=1}^r \prod_{j=1}^s t_{ij}^{x_{ij} / x_{ij}!} \right\} du_1 du_2 \dots du_s.$$

From the fact that (Dirichlet's integral):

$$\int_{t_1 \geq 0} \int_{t_2 \geq 0} \dots \int_{t_n \geq 0} f(t_1 + t_2 + \dots + t_n) t_1^{\alpha_1-1} t_2^{\alpha_2-1} \dots t_n^{\alpha_n-1} dt_1 dt_2 \dots dt_n \\ = \left(\prod_i \Gamma(\alpha_i) / \Gamma(\sum_{i=1}^n \alpha_i) \right) \int_0^1 f(w) w^{\sum_{i=1}^n \alpha_i - 1} dw$$

(15) can be written, after some straightforward manipulation, as

$$\sum_{x_{11}} \sum_{x_{12}} \dots \sum_{x_{rs}} \frac{\prod_{i=1}^r [\sum_{j=1}^s x_{ij}] \prod_{j=1}^s [\sum_{i=1}^r x_{ij}]}{\prod_{i=1}^r \prod_{j=1}^s x_{ij}!} \frac{\prod_{i=1}^r \prod_{j=1}^s t_{ij}^{x_{ij}}}{x_{ij}!}$$

which is $F_{1;r,s}(\cdot)$ as defined in (4).

A.2 A SYSTEM OF DIFFERENTIAL EQUATIONS

Writing (4) as: $\sum_{x_{11}} \sum_{x_{12}} \dots \sum_{x_{rs}} g(x_{11}, x_{12}, \dots, x_{rs}) \prod_{i=1}^r \prod_{j=1}^s t_{ij}^{x_{ij}}$

$$\text{where } g(x_{11}, x_{12}, \dots, x_{rs}) = \frac{\prod_{i=1}^r [\sum_{j=1}^s x_{ij}] \prod_{j=1}^s [\sum_{i=1}^r x_{ij}]}{\prod_{i=1}^r \prod_{j=1}^s x_{ij}!}$$

it follows that:

$$(16) \quad \frac{(a_\ell + \sum_{j=1}^s x_{\ell j} - 1)(b_m + \sum_{i=1}^r x_{im} - 1)}{(c + \sum_{i=1}^r \sum_{j=1}^s x_{ij} - 1) x_{\ell m}} g(x_{11}, x_{12}, \dots, x_{\ell m}^{-1}, \dots, x_{rs})$$

$$= g(x_{11}, x_{12}, \dots, x_{\ell m}, \dots, x_{rs})$$

Multiplying both sides of (16) by $\prod_{i=1}^r \prod_{j=1}^s t_{ij}^{x_{ij}}$ and summing over all values of $x_{11}, \dots, x_{rs}$ it follows after a considerable amount of algebra that (16) leads to the following system of differential equations which are satisfied by $F_{1;r,s}(\cdot)$:

$$(1-t_{\ell m}) \sum_{i=1}^r \sum_{j=1}^s t_{ij} \frac{\partial^2 F}{\partial t_{\ell m} \partial t_{ij}} + \sum_{i \neq \ell}^r \sum_{j \neq m}^s t_{ij} \frac{\partial^2 F}{\partial t_{\ell m} \partial t_{ij}} \quad (a)$$

$$- \sum_{i \neq \ell}^r \sum_{j \neq m}^s t_{ij} t_{\ell m} \frac{\partial^2 F}{\partial t_{\ell m} \partial t_{ij}} + (c - (a_{\ell} + b_m + 1)t_{\ell m}) \frac{\partial F}{\partial t_{\ell m}} - a_{\ell} \sum_{i \neq \ell}^r t_{im} \frac{\partial F}{\partial t_{im}} - b_m \sum_{j \neq m}^s t_{\ell j} \frac{\partial F}{\partial t_{\ell j}} - a_{\ell} b_m F = 0,$$

for $\ell = 1, 2, \dots, r$ and $m = 1, 2, \dots, s$ and where $\sum_{i=1}^r \sum_{j=1}^s$ indicates summation over i and j where the following combinations of i and j are possible:

- $i = \ell, j = m$
- $i = \ell, j \neq m$
- $i \neq \ell, j = m.$

A.3 THE FMGF WHEN $n - n_{h.} - n_{.k} > 0$

In writing down the expression (3) the condition $n_{h.} + n_{.k} - n \geq 0$ was necessary

$$f_{hk}! = (n_{h.} + n_{.k} - n + \sum_{i=1}^{h-1} \sum_{j=1}^{k-1} f_{ij})! = (n_{h.} + n_{.k} - n)! (n_{h.} + n_{.k} - n + \sum_{i=1}^{h-1} \sum_{j=1}^{k-1} f_{ij})^{\sum_{i=1}^{h-1} \sum_{j=1}^{k-1} f_{ij}}$$

If $n - n_{h.} - n_{.k} > 0$, the pgf can be written as

$$(17) n_{h.}! n_{.k}! \sum \dots \sum \left[\frac{\sum_{i=1}^{h-1} n_{i.} \prod_{j=1}^{k-1} n_{ij}^{\sum_{i=1}^{h-1} \sum_{j=1}^{k-1} f_{ij}}}{(\sum_{i=1}^{h-1} \sum_{j=1}^{k-1} f_{ij} - (n - n_{h.} - n_{.k}))!} \prod_{i=1}^{h-1} \prod_{j=1}^{k-1} \frac{t_{ij}^{f_{ij}}}{f_{ij}!} \right]$$

where $\Sigma \dots \Sigma$ indicates summation over all values of, $f_{11}, \dots, f_{(h-1)(k-1)}$ under the condition $\sum_{i=1}^{h-1} \sum_{j=1}^{k-1} f_{ij} \geq n - n_h - n_k$.

It is important to note that $\sum_{i=1}^{h-1} \sum_{j=1}^{k-1} f_{ij}$ cannot be zero, although any particular $f_{ij} \geq 0$, and that the sum inside the square brackets must be equal to $n!/(n_h!n_k!)$ since the total probability should be equal to one.

Substituting $t_{ij} = 1 + \alpha_{ij}$, $i=1, 2, \dots, h-1$; $j=1, 2, \dots, k-1$ the fmgf follows as:

$$\frac{n_h!n_k!}{n!} \Sigma \dots \Sigma \frac{\prod_{i=1}^{h-1} \prod_{j=1}^{k-1} n_i^{(\sum_{j=1}^{k-1} f_{ij})}}{(\sum_{i=1}^{h-1} \sum_{j=1}^{k-1} f_{ij} - (n - n_h - n_k))!} \prod_{i=1}^{h-1} \prod_{j=1}^{k-1} \frac{(1 + \alpha_{ij})^{f_{ij}}}{f_{ij}!}$$

where $\Sigma \dots \Sigma$ is defined in (17), so that the coefficient of

$$\frac{\alpha_{11}^{y_{11}} \alpha_{12}^{y_{12}} \dots \alpha_{(h-1)(k-1)}^{y_{(h-1)(k-1)}}}{y_{11}! y_{12}! \dots y_{(h-1)(k-1)}!}$$

is given by

$$\frac{n_h!n_k!}{n!} \prod_{i=1}^{h-1} \prod_{j=1}^{k-1} n_i^{(\sum_{j=1}^{k-1} y_{ij})} \prod_{j=1}^{k-1} \prod_{i=1}^{h-1} n_j^{(\sum_{i=1}^{h-1} y_{ij})} \Sigma \dots \Sigma \left[\frac{\prod_{i=1}^{h-1} \prod_{j=1}^{k-1} n_i^{(\sum_{j=1}^{k-1} (f_{ij} - y_{ij}))} \prod_{j=1}^{k-1} \prod_{i=1}^{h-1} n_j^{(\sum_{i=1}^{h-1} (f_{ij} - y_{ij}))}}{(\sum_{i=1}^{h-1} \sum_{j=1}^{k-1} (f_{ij} - y_{ij}) - (n - n_h - n_k - \sum_{i=1}^{h-1} \sum_{j=1}^{k-1} y_{ij}))!} \prod_{i=1}^{h-1} \prod_{j=1}^{k-1} 1/(f_{ij} - y_{ij})! \right]$$

where $\Sigma \dots \Sigma$ indicates summation over all values of

$f_{11}^{-y_{11}}, \dots, f_{(h-1)(k-1)}^{-y_{(h-1)(k-1)}}$ under the condition:

$$\sum_{i=1}^{h-1} \sum_{j=1}^{k-1} (f_{ij}^{-y_{ij}}) \geq n - n_{h.} - n_{.k} - \sum_{i=1}^{h-1} \sum_{j=1}^{k-1} y_{ij}$$

The sum of the series in the square brackets, using the note following (17) is equal to

$$(n - \sum_{i=1}^{h-1} \sum_{j=1}^{k-1} y_{ij})! / (n_{h.}! n_{.k}!) \dots$$

The coefficient of $\prod_{i=1}^{h-1} \prod_{j=1}^{k-1} (\alpha_{ij}^{y_{ij}} / y_{ij}!)$ in the fmgf reduces to:

$$\begin{aligned} & \frac{n_{h.}! n_{.k}!}{n!} \prod_{i=1}^{h-1} \frac{(\sum_{j=1}^{k-1} y_{ij})!}{n_{i.}!} \prod_{j=1}^{k-1} \frac{(\sum_{i=1}^{h-1} y_{ij})!}{n_{.j}!} \cdot \frac{(n - \sum_{i=1}^{h-1} \sum_{j=1}^{k-1} y_{ij})!}{n_{h.}! n_{.k}!} \\ &= \frac{\prod_{i=1}^{h-1} \frac{(\sum_{j=1}^{k-1} y_{ij})!}{n_{i.}!} \prod_{j=1}^{k-1} \frac{(\sum_{i=1}^{h-1} y_{ij})!}{n_{.j}!}}{(n - \sum_{i=1}^{h-1} \sum_{j=1}^{k-1} y_{ij})!} \end{aligned}$$

so that the fmgf is again given by:

$$F(-n_{1.}, -n_{2.}, \dots, -n_{(h-1).}, -n_{.1}, -n_{.2}, \dots, -n_{.k}, -n; -\alpha_{11}, -\alpha_{12}, \dots, -\alpha_{(h-1)(k-1)})$$

which is the same function obtained in (9) for the case

$$n_{h.} + n_{.k} - n \geq 0.$$